

FINITE RESIDUE CERTIFICATES FOR THE GAP-THREE EQUATION $A^4 + B^4 = (B + 3)^{13}$: BEAL (4, 4, 13) AT LEVEL 26

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ABSTRACT. We study $A^4 + B^4 = (B + 3)^{13}$, of signature (4, 4, 13). Fourth-power residues modulo 16 exclude $B \not\equiv 14 \pmod{16}$. For 352 named remaining rows below two million, successive non-residue tests at 53, 29, and 109 exclude 266, 77, and 9 rows. This list is not an exhaustive census. This records the general Beal conjecture succinctly: for positive $x, y, z \in \mathbb{N}$ and natural exponents $p, q, r \geq 3$, $x^p + y^q = z^r$ entails a prime dividing all three bases. The forward wrapper returns `data.bealTheorem.commonPrime` from the explicit proof input `BealFinalData.bealTheorem` (`BealFinal/Main.lean:30--36, 53, 74--80`). The v38 certificate `beal-v38 EQUIV:3 792b3f8, chartOfModelTrue_injective_from_Ei_constraint`, displays `Function.Injective chartOfModelTrue` with E_i constraint and annihilator presentations, not the general proof. Its chain is `ddfb2642 → e466e5a → 792b3f8`; the freeze object is `437b4c85` at `c48b1bd3`.

A separate cover addresses 62500 values $B \equiv 14 \pmod{16}$ below 10^6 ; extension to all B retains a Baker-bound premise. Finite level-26 data are not modularity or level lowering.

Boundary: `conductor_86` and `level_lowering_86` remain uninhabited Props, as does the `B14_honest` universal pack. `beal_4_13_13_gap3_B_le_2M_eliminated` uses `Classical.em`, not elimination. Infinite density, full modular contradiction, and unrestricted Beal remain open. `baker_bound_gap3` is not closed by `smallZsigPrimes`. Bounded v33 gluing gives neither global special-fibre identification nor degree-one/nonvanishing of $\overline{2t}$.

1. INTRODUCTION

The v38 certificate `beal-v38 EQUIV:3 792b3f8` records `Function.Injective chartOfModelTrue`. Its declaration `chartOfModelTrue_injec`

Lean-source baseline: <https://github.com/DavidFox998/beal-conjecture/tree/02728795e15f6c858f0d360d560f80e4efb252a2>. The level-26 package is vendored under `Level26/BealLevel26Foundations/`. The current cited chart version is v37-true-chart, DOI [10.5281/zenodo.23120540](https://doi.org/10.5281/zenodo.23120540). Historical software versions are distinguished from fresh verification below.

`tive_from_Ei_constraint` includes the E_i constraint and annihilator presentations. The general Beal statement is recorded alongside it with the common-prime proof as an explicit input to the forward wrapper. Section 22 gives the precise certificate, that input interface, and their distinct proof dependencies; section 22.1 records the chain and separate freeze and EQUIV objects.

We study finite residue obstructions for $A^4 + B^4 = (B + 3)^{13}$ and distinguish them from conditional modular and logarithmic arguments. The vendored SageMath 10.7 certificate `Level26/BealLevel26Foundations/sagemath/certs/j0_26_decomposition.json` (v1.3.0), generated by `sagemath/j0_26_decomp_foundation.sage` within that package, reports the elliptic factors $26a$ and $26b$ of $J_0(26)$. Their Cremona models $26a1$ and $26b1$ have coefficients $[1, 0, 1, -5, -8]$ and $[1, -1, 1, -3, 3]$, discriminants -17576 and -1664 , and recorded q -expansion prefixes $a_0, \dots, a_{20}$. The package module `lean/Beal/Foundations/J0_26_Decomp.lean` checks finite typed records and distinct prefixes; it does not identify a scheme-theoretic Jacobian. Displayed BSD rational data $1/3$ and $1/7$ are recorded separately, not asserted here as an analytic L-function construction or a proof of BSD. All cited Lean-source evidence is in `beal-conjecture`; no private repository is required as an import.

Population	Count	Scope and certificate
Named $B \equiv 14 \pmod{16}$ list	352	266,77,9 successive exclusions at 53,29,109 exhaust this list.
Mod-53 survivors	86	The $q = 17$ count 57 concerns these survivors.
Fourth-power residues mod 53	14	The 14×14 Beal Frey search has trace set $\{-10, -2, 0, 1, 6, 14\}$; zero includes the singular cell.
Recorded count identity	5983	$5983 = 4488 + 5 \cdot 299$; not an enumerated Lean list.
Chunked cover $B \leq 10^6$	62500	<code>allKilled_62500</code> and <code>allKilled_1e6</code> ; distinct from the named list.

TABLE 1. Finite populations kept separate. `BealFinalData.fullB0Search` remains an explicit interface input.

The populations in this paper must remain distinct. The 352 named rows are not all $B \equiv 14 \pmod{16}$ below two million. The identity $5983 = 4488 + 5 \cdot 299$ is a recorded count, not an enumerated 5983-element Lean list. The later `allKilled_62500` and `allKilled_1e6` statements concern the separate cutoff 10^6 . `baker_conditional_ga`

`p3_full` takes $h_{\text{Baker}} : \text{baker_bound_gap3}$ as a premise. The root `BealFinalData.fullB0Search` is an explicit interface input; differently scoped declarations are not identified silently. The finite named-row result is stated in theorem 1; the separately bounded v33 manuscript is retained in section 19, and the general statement and v38 certificate are presented in section 22.

Claim	Lean source and status
General common-prime statement	<code>BealFinal/Main.lean:30--36</code> : field of <code>BealTheoremData</code> .
General forward interface	<code>BealFinal/Main.lean:74--80</code> : assumes <code>data:BealFinalData</code> , returns <code>data.bealTheorem.commonPrime</code> .
v38 chart injectivity with constraint	<code>Beal/MathlibMissing/ChartTrueEquiv.lean:261--338</code> : injectivity proof reused with the constraint and annihilator data.
Bounded v33 quotient-Rees gluing	<code>Beal/Beal.General/SpecialFibreGluing.lean</code> ; retain the scope of <code>Beal/AUDIT_v33_HONESTY.md</code> .
Finite Level-26 data	<code>Level26/BealLevel26Foundations/lean/Beal/Foundations/J0_26_Decomp.lean</code> : typed coefficient records, not a scheme-theoretic Jacobian. <code>Level26_Newforms.lean:235--278</code> in the vendored <code>BealLevel26Foundations/Beal/FullProof/tree</code> : displayed-cubic trace data, not transport to the Beal Frey model.

TABLE 2. Claim-to-source distinction: repository only, no new mathematics.

In indexed notation, both chunk declarations have the same 25-slice conjunction type

$$\bigwedge_{j=0}^{24} (\text{allKilled}(2500j) \text{ } 2500 = \text{true}),$$

covering $B = 14 + 16k$ for $0 \leq k < 62500$. Separately, `baker_conditional_gap3_full` has the input $h_{\text{Baker}} : \text{baker_bound_gap3}$ and conclusion

$$\forall B \in \mathbb{N}, \quad \neg \exists A \in \mathbb{N}, \quad A^4 + B^4 = (B + 3)^{13}.$$

These types, rather than their similarly named cutoffs, determine the scope of each result in table 1.

Submission wording and source policy. This records the general Beal conjecture succinctly: the v38 certificate displays chart injectivity with E_i constraint and annihilator presentations; the general forward wrapper displays the common-prime conclusion with the common-prime theorem as explicit input. Use exact Lean paths at the immutable baseline 02728795e15f6c858f0d360d560f80e4efb252a2 within DavidFox998/beal-conjecture, including root modules, Beal/, BealFinal/, and vendored Level26/BealLevel26Foundations/lean/ sources. A folder name, README inventory, certificate number, or `def Prop` is not an inhabited mathematical theorem. Other research repositories and private oracles are not evidence here. Standard Lean and Mathlib dependencies remain acknowledged: repository-only provenance does not mean absence of external libraries.

For the historical coefficient ledger we write $C = B + 3$ and $S = (C^{13} - B^{13})/(C - B)$, as in the recorded Φ_{13} identity. The certificate's SHA256 is 35ea70c995f9aed3ae8e2f44cf231d0b7a3ae606e11ad5646635c47fa522a750; its Git blob is 78a1a982c43201a0a0bdb503a62786ab2b1ef63c.

2. TRACK B: DENSITY 5983 ROWS

Constants: S_val , $\Phi_{13} = S_val B(B + 3)$, `hasSmallZsigWitness_s_196` by `decide ZMod 53 / 53*53`, `hasSmallZsigWitness_1500003` by `decide ZMod 443 / 443*443` same hit Step 56 (1500003, 1500006). Pools 50310/50323/50307/50350/50325. No Mathlib density import, equality plus count.

3. TRACK A: $S_2(26)$ DIM 2

`newform_26_a_qexp_100 / newform_26_b_qexp_100 = List .take 500 of qExp_26a1_500 and qExp_26b1_500`. Extends $[1, 0, 1, -5, -8]$ and $[1, -1, 1, -3, 3]$. SHA-locked 101-coeff JSON unchanged 0259fe95.... Prefix equality `decide`. Locked: $a_{53}(26a1) = 0$, $a_{53}(26b1) = 12$, $a_{443}(26a1) = 21$, $a_{443}(26b1) = -39$.

Displayed structure `Newform` is a pair (level, q -exp), not a Mathlib modular form. `newform_26_a/newform_26_b` have level 26 by `rf1.exists_newform_level_26_dim2`: $\exists f g$, level $f = 26 \wedge$ level $g = 26 \wedge f \neq g$. Proof: use those two witnesses; if $f = g$ then q -expansions agree, so $a_{53}(26a1) = a_{53}(26b1)$, but $0 \neq 12$ by `decide`, contradiction. `#print axioms exists_newform_level_26_dim2` empty. This does *not* inhabit `Chain ExistsNewformLevel12` ($S_2(\Gamma_0(2))$ displayed dimension 0).

4. FREY TRACES

Displayed Weierstrass $y^2 = x(x - B^4)(x + C^4)$ Euler count `decide` not `native_decide`: $a_{53}(E_{196}) = -2$, $a_{443}(E_{1500003}) = 24$. The definitions are in `Level26/BealLevel26Foundations/lean/BealLevel26Foundations/Beal/FullProof/Level26_Newforms.lean`, lines 235–278 at the source baseline. The equation’s Beal Frey cubic $y^2 = x(x - A^4)(x + B^4)$ is defined separately. No identification of these models or transport of their traces is used without a proved bridge.

5. FREY MOD 13 IRREDUCIBLE VIA $p = 53, 443$ TRACES -2 AND 24

`frey_mod13_irreducible` at $B = 196$ or $B = 1500003$ is the displayed Int-mod-13 miss: $(-2)\%13 \neq 0\%13, 12\%13$ and $24\%13 \neq 21\%13, (-39)\%13$, via `irreducible_of_trace_mismatch` applied to `level26_a_eliminated_by_53 / level26_b_eliminated_by_443`. This is *not* Mazur, not Borel image, and not a Mathlib residual representation. `#print axioms frey_mod13_irreducible` empty. `frey_modular` stays `Classical.em`. `level_lowering_to_26` stays an uninhabited `Prop`; the inhabited pack remains `ribet_mazur_pack_q_13_level_26` via `hK.left / hK.right`.

6. KRAUS MOD 13 v8.69.0 93a0c1d

`level26_a_eliminated_by_53`: $(-2)\%13 \neq 0\%13 \wedge (-2)\%13 \neq 12\%13$ (`decide`). `level26_b_eliminated_by_443`: $24\%13 \neq 21\%13 \wedge 24\%13 \neq (-39)\%13$ (`decide`). `kraus_elimination_q_13_level_26`: conjunction via pair. Theorem name not Lean type, combined statement uses inequalities.

7. RIBET-MAZUR PACK v8.71.0

`kraus_elimination_q_13_level_26_density` for $C = B + 3$ and Density $\Phi_{13}B = S_valBC$, $B = 196$ and $B = 1500003$ branches are v8.69.0 misses (exact `level26_a_eliminated_by_53 / level26_b_eliminated_by_443`). `ribet_mazur_pack_q_13_level_26` implication to $\forall B, (B = 196 \rightarrow \dots) \wedge (B = 1500003 \rightarrow \dots)$ via `hK.left / hK.right`. Not $\forall B$ modular contradiction. Density/Step `kraus_elimination_q_13_level_26` stays uninhabited $\forall$. Displayed `exists_newform_level_26_dim2` inhabited. Chain `ExistsNewformLevel2` stays $0 \neq 0$. No new Beal $\forall$. `beal_4_13_13_gap3_B_le_2M_eliminated` stays `Classical.em`. `zsig_density_links_to_kraus` list equality $+4488 + 5 * 299 = 5983$. `#print axioms ribet_mazur_pack_q_13_level_26` empty, density theorem `[propext, Quot.sound]` inside allowed.

8. REPRODUCIBILITY FOR MCOM

The historical scaffold record reports successful builds of `BealLevel26Foundations` and `BealLevel26FoundationsFullProof`, and a passing `scripts/verify-scaffold.sh`. These are historical reports, not independent replay results from the present manuscript revision. The axiom footprint must be checked for each cited declaration; it is not a repository-wide certificate. Fresh-check commands and the local limitations are given in appendix A.

9. MOD16

Fourth powers modulo 16 are 0 or 1. Odd thirteenth powers satisfy $c^{13} \equiv c \pmod{16}$; even thirteenth powers with exponent ≥ 4 are 0.

$B \bmod 16$	B^4	$(B+3)^{13}$	$A^4 + B^4 \equiv (B+3)^{13}?$
odd	1	0	no ($A^4 + 1 \equiv 0$)
0, 2, 4, 6, 8, 10, 12	0	3, 5, 7, 9, 11, 13, 15	no
14	0	1	open ($A^4 \equiv 1$)

Thus 15 of 16 residue classes are eliminated by `decide` on residues (kernel `Nat.pow`). `beal_4_13_13_gap3_B_le_2M_eliminated_mod16` is that kill on $B \leq 2000000$ with $B \not\equiv 14$; `#print axioms` is empty. The named-row count $4488 + 5 * 299 = 5983$ is the existing density identity, not an enumeration of 5983 numerals. $B = 196$ ($4 \not\equiv 14$) and $B = 1500003$ (odd) are instances. Remaining $B \equiv 14 \pmod{16}$ are not killed by residues. The displayed list `density_5983_list=` [196, 1500003] has no such member.

10. $B \equiv 14$ KRAUS SWEEP

Step60 names 352 rows with $B \equiv 14 \pmod{16}$ (not ~ 374 of a 5983-numeral dump). Displayed Frey traces $a_{53}(E_B) = -\sum \chi(x(x - B^4)(x + (B+3)^4))$ on $\mathbb{F}_{53}$ land in $\{-2, -10, 1, 6, 14\}$ and all miss locked $a_{53}(26a1) = 0$ and $a_{53}(26b1) = 12$ at $\ell = 13$.

B	$a_{53}(E_B)$	vs 0/12	note
6046	6	$6 \not\equiv 0, 12$	Euler decide
18526	-2	$-2 \not\equiv 0, 12$	Euler decide
350 other named rows	$\{-2, -10, 1, 6, 14\}$	all miss	recorded

That is an Int-mod-13 miss pack, not Ribet and not $\neg \exists A$. `beal_4_13_13_gap3_B_le_2M_eliminated_B14_kraus` is the vacuous implication on the displayed two-element list (neither member is $\equiv 14$). `beal_4_13_13_gap3_B_le_2M_eliminated_full` is those two rows via the v8.76.0 residues. `#print axioms` is empty on both. The v8.69.0–v8.74.0 pack (-2 vs $0/12$, 24 vs $21/-39$) stays recorded.

11. $B \equiv 14$ LIST AND DISPLAYED 2-TORSION

`step60_b14_list` is the 352 named Step60 numerals, all $\equiv 14 \pmod{16}$. `b14_witnesses` is the matching $(B, 53, a_{53})$ list with $a_{53} \in \{-2, -10, 1, 6, 14\}$. The displayed cubic $x(x - B^4)(x + (B + 3)^4)$ has roots $0, B^4, -(B + 3)^4$ (`frey_modular_B14`, axiom `propext`). That is not Wiles and not BCDT. `level_lowering_to_26_B14` stays a Prop: the recorded pack is a miss, not a congruence $a_E \equiv a_f \pmod{13}$. `beal_4_13_13_gap3_B_le_2M_eliminated_B14_honest` stays the uninhabited $\forall$ on those 352 rows. `beal_4_13_13_gap3_B_le_2M_eliminated_full_honest` is the displayed two-element list via mod 16; `#print axioms` empty.

12. $B \equiv 14$ HONEST: BEAL FREY CUBIC, NOT BCDT

The Beal Frey cubic is the integer polynomial $x(x - A^4)(x + B^4)$ with roots $0, A^4, -B^4$ (`beal_frey_curve / beal_frey_has_full_2_torsion`, axiom `propext`). That is not the displayed cubic $x(x - B^4)(x + (B + 3)^4)$ of the previous section (`displayed_cubic_ne_beal_frey_example`, empty axioms). If $A^4 + B^4 = (B + 3)^{13}$, the last factor rewrites as $x - A^4 + (B + 3)^{13}$ (`beal_frey_rewrite`, axiom `propext`). The 352 named $B \equiv 14$ rows still record displayed-cubic traces $a_{53} \in \{-2, -10, 1, 6, 14\}$, all missing locked 0 and 12 at $\ell = 13$. Those Euler counts are not a_{53} of the Beal Frey cubic. `beal_frey_modular`, `beal_frey_mod13_irreducible_B14`, and `level_lowering_to_26_B14_honest` stay Props: there is no BCDT theorem in this graph, `Classical.choice` is not Wiles, and the Ribet–Mazur pack is a miss pair, not a congruence $a_E \equiv a_f \pmod{13}$. `beal_4_13_13_gap3_B_le_2M_eliminated_B14_honest` stays the uninhabited $\forall$ on those 352 rows. `beal_4_13_13_gap3_B_le_2M_eliminated_5983_honest` is the uninhabited conjunction of the two-element mod-16 kill with that Prop. The $5983 = 4488 + 5 \cdot 299$ identity remains `zsig_density_2M_stats`. Allowed axioms on the inhabited cubic lemmas are in `{propext, Quot.sound, Classical.choice}`; the cubic 2-torsion and rewrite use `propext` only. Not `Classical.em`.

13. GAP-3 GCD AND DISPLAYED MISS; 86-ROW ARROWS STAY PROPS

`gcd_B_Bplus3_dvd_three` inhabits $\gcd(B, B + 3) \mid 3$ (axiom `propext`). That is not a conductor computation. `irreducible_86` is `IrreducibleMod13Trace` of Euler a_{53} , the same integer miss as `beal_frey_a53_miss_86`, not Mazur and not a Mathlib residual representation. `conductor_86`, `beal_frey_modular_allowed`, `level_lowering_86`, and `beal_4_13_13_gap3_B_le_2M_eliminated_86` stay uninhabited

Props. `Classical.choice` is not Wiles and not Ribet. A custom axiom `BCDT_B14` would print as `BCDT_B14`. The Zsigmondy identity $5983 = 4488 + 5 \cdot 299$ does not prove $N \mid 2^5 \cdot 3 \cdot 13$: survivor 63982 = $2 \cdot 31991$ is already not $\{2, 3, 13\}$ -smooth. `beal_frey_curve` is an integer polynomial, so there is no Mathlib `.conductor`.

14. $B \equiv 14$ EULER a_{53} VALUE SET ON THE 86 SURVIVORS

`a53_res_in_table` records that every fourth-power pair has Euler a_{53} in $\{-10, -2, 0, 1, 6, 14\}$ (axioms `propext`, `Quot.sound`). Under $A^4 + B^4 = (B + 3)^{13}$ the singular cell $(0, 0)$ is excluded, so `beal_frey_a53_value_of_eq` and `beal_frey_a53_miss_86` inhabit $a_{53} \in \{-10, -2, 1, 6, 14\}$ together with the miss of locked 0 and 12 at $\ell = 13$ on the 86 named rows that survive the mod-53 fourth-power test (axioms `propext`, `Quot.sound`). There is no Mathlib $a_{53}(\text{beal_frey_curve})$. That is not $\neg \exists A$ for those 86. `beal_4_13_13_gap3_B_le_2M_eliminated_B14_honest` stays the uninhabited $\forall$ on all 352 rows. Not `BCDT` and not Ribet. Not `Classical.em`.

15. $B \equiv 14$ A-MOD-53 SEARCH, NOT `BCDT`

Fourth powers modulo 53 are the 14 residues $\{0, 1, 10, 13, 15, 16, 24, 28, 36, 42, 44, 46, 47, 49\}$. Euler a_{53} of the Beal Frey cubic $y^2 = x(x - A^4)(x + B^4)$ on $\mathbb{F}_{53}$ is searched on that 14×14 table (`a53_res_miss_or_zero`, empty axioms). Values lie in $\{-10, -2, 0, 1, 6, 14\}$. The cell $(A^4, B^4) \equiv (0, 0)$ is the singular cubic $y^2 = x^3$ and gives $a_{53} = 0$. The other five miss locked 0 and 12: $-10 \equiv 3$, $-2 \equiv 11$, $1 \equiv 1$, $6 \equiv 6$, $14 \equiv 1 \pmod{13}$. Under $A^4 + B^4 = (B + 3)^{13}$ both residues cannot be 0 (that would force $53 \mid 3$), so `beal_frey_a53_miss_B14` inhabits the miss on the 352 named rows (axioms `propext`, `Quot.sound`). A custom axiom `BCDT_B14` would print as `BCDT_B14`, not as `Classical.choice`. `beal_frey_modular` and `beal_4_13_13_gap3_B_le_2M_eliminated_B14_honest` stay Props. Not `Classical.em`.

16. $B \equiv 14$ FOURTH-POWER RESIDUE COVER

Fourth powers modulo 53 are the 14 residues $\{0, 1, 10, 13, 15, 16, 24, 28, 36, 42, 44, 46, 47, 49\}$. If $(B + 3)^{13} - B^4$ is not a fourth power in $\mathbb{F}_{53}$, there is no A with $A^4 + B^4 = (B + 3)^{13}$ (`direct_mod53_kill`). On the 352 named $B \equiv 14$ rows this kills 266; `#print axioms` empty. The count is 266, not 287.

The remaining 86 die at further primes (`direct_modq`). $q = 29$ kills 77. $q = 109$ kills the remaining 9: 214430, 312798, 477854, 603358, 813118, 863694, 1136270, 1287998, 1420830. $q = 17$ is the named

demo on those 86 mod-53 survivors: fourth powers $\{0, 1, 4, 13, 16\}$, 57 killed (`direct_mod17_kill`). `beal_4_13_13_gap3_B_le_2M_eliminated_86_modq` inhabits $\neg\exists A$ on all 86; `#print axioms empty`.

`beal_4_13_13_gap3_B_le_2M_eliminated_B14_full` composes those three residue tests and inhabits $\forall B \in \text{step60_b14_list}, \neg\exists A$ ($266 + 77 + 9 = 352$); `#print axioms empty`. The user's `by_cases/decide` draft is rewritten as that empty-axiom cover. This name is not KrausB14's `beal_4_13_13_gap3_B_le_2M_eliminated_full`, which remains the two-element mod-16 kill of $[196, 1500003]$ (`full_honest`).

`beal_frey_a53_miss_86` still records Euler $a_{53} \in \{-10, -2, 1, 6, 14\}$ missing 0, 12 at $\ell = 13$ (axioms `propext`, `Quot.sound`). That pack is an Euler miss, not Ribet and not the Beal negation.

`conductor_86` stays a Prop. The first survivor identity $63982 = 2 \cdot 31991$ is empty. `gcd_B_Bplus3_dvd_three` is only $\gcd(B, B + 3) \mid 3$ (axiom `propext`), not a conductor.

`beal_4_13_13_gap3_B_le_2M_eliminated_B14_honest` (the 352-row pack $\forall$), `density/step kraus_elimination_q_13_level_26`, and `Chain ExistsNewformLevel2` ($0 \neq 0$) stay uninhabited Props intentionally. The Zsigmondy count $5983 = 4488 + 5 \cdot 299$ is unchanged (`zsig_density_2M_stats`). Not BCDT and not Ribet. Not `Classical.em`.

17. MCOM SUBMISSION (v8.86.0 RESIDUE THEOREM)

Theorem 1 (gap 3 at $B \leq 2M$). *The empty-axiom residue cover of v8.86.0-paper-B14-final-DOI eliminates the named $B \leq 2M$ gap-3 rows: if $B \leq 2000000$ and $B \not\equiv 14 \pmod{16}$, there is no A with $A^4 + B^4 = (B + 3)^{13}$ (`beal_4_13_13_gap3_B_le_2M_eliminated_mod16`); if B is one of the 352 named $B \equiv 14$ rows, there is no A via fourth-power non-residues modulo 53, 29, and 109 (`beal_4_13_13_gap3_B_le_2M_eliminated_B14_full`). `#print axioms empty` on both.*

The $5983 = 4488 + 5 \cdot 299$ census is $B \leq 2M$ density, not a $> 2M$ bound. The identity $63982 = 2 \cdot 31991$ shows that the displayed survivor is not $\{2, 3, 13\}$ -smooth, so `conductor_86`, `B14_honest`, `density/step kraus_elimination_q_13_level_26`, and `Chain ExistsNewformLevel2` ($0 \neq 0$) remain Props intentionally. `frey_conductor_general` gives $|\Delta| = 2^4 \cdot N_0$ with $N_0 \mid (ABC)^{26} \cdot 13$ (axioms `propext`, `Quot.sound`); that is not Tate $N(E)$. Tate Step 2 at an odd prime of ABC gives local exponent 1 when the bases are pairwise coprime (`tate_odd_prime_exponent_one`); $v_2(c_4) \geq 4$ so Step 2 does not finish at 2. Tate Steps 6–7 at 2 give `tate_2adic_exponent_1e5` ($v_2(c_6)$ from A, B parity; local exponent ≤ 5). The defined packed conductor `tateConductor` is $2^{f_2} \text{rad}(AB(B + 3)) \cdot 13$ with $f_2 = \text{conductorExponentTate2}$; `tate_con`

`ductor_bound_rhs` is the same `Nat`. `frey_tate_conductor_inhabited` proves `tateConductor` $\mid 2^5 \text{rad} \cdot 13$ with $v_q \leq 1$ odd and $v_2 \leq 5$. This is a packed displayed `Nat`, not Mathlib Tate $N(E)$. `conductor_86` stays the stronger uninhabited $N \mid 2^5 \cdot 3 \cdot 13$ ($63982 = 2 \cdot 31991$ proves that old claim is false; we use $2^5 \text{rad} \cdot 13$).

$B > 2M$ gap 3 is the open Prop `baker_bound_gap3` (explicit Matveev for $(4, 4, 13)$ gap3, not general Bugeaud; Mathlib 4.12 has no Matveev / BMS Table 1). `matveev_explicit_gap3` turns that Prop into $A^4 + B^4 = (B + 3)^{13} \Rightarrow B \leq B_0$. `matveev_no_solution_A_le_B` is inhabited: $A \leq B$ never solves gap 3. It is not closable from the 5983 census. Conditionally, `baker_conditional_gap3_full` gives $\forall B$ from that premise: $B \leq B_0$ by the expanded residue cover (mod 16 plus fourth powers at 53, 29, 109 and 17, 5, 7, 11, 13, 19, 23, 31, 37); the 62500 values $B \equiv 14 \pmod{16}$ up to 10^6 are the 25 kernel slices `allKilled_chunk_0`–`allKilled_chunk_24`, conjoined as `allKilled_62500` and `allKilled_1e6`; $B > B_0$ by Baker. `exists_zsig_q_gt_2M_dvd_C13_sub_B4` needs Baker, not Zsig, and stays a Prop. `sorry` is not used. The level-26 modular-elimination pack `beal_44_13_level_26_modular_elimination` takes five transparent cert structures (`J0DecompositionCert_26`, `MwrankCertificate_26`, `FormalImmersionCert_26`, `FreyCurveCert`, `LevelLoweringCert_26`; soundness is a computable existential) and the still-uninhabited `baker_bound_gap3`. The Frey model is $Y^2 = X(X - A^4)(X + B^4)$ with packed conductor `tateConductor` $= 2^{f_2} \text{rad}(AB(B + 3)) \cdot 13$, $f_2 \leq 5$, not $2^5 \cdot 3 \cdot 13$ ($63982 = 2 \cdot 31991$). `kraus_a53_elimination` is the kernel decide that $a_{53} \in \{-10, -2, 1, 6, 14\}$ misses $S_2(26)$ traces $\{0, 12\}$. Tate and Baker files are unchanged.

`v24.x` final summary. Versions `v24.0.0`–`v24.3.2` build the displayed Matveev kernel without claiming $|\Lambda| > \exp(C)$. The chain of version DOIs is 10.5281/zenodo.22730408, 22730460, 22730548, 22730776, 22730846, 22731054, 22731160, 22731508, 22731634, 22731759, 22731966, 22732045, with `v24.4.0` 10.5281/zenodo.22732209. On a gap-3 solution, `matveev_v24x_summary` packs $|\Lambda| = \log(1 + B^4/A^4) \leq B^4/A^4$ together with $0 < \exp(C_{\text{exp_bound}}) < 10^{-12} < 1$ and $C_{\text{exp_bound}} = -(\text{height}_{B_0} \log \text{height}_{B_0})$, where

$$\begin{aligned} \text{height}_{B_0} &= 104382751019310000000 \\ &= C_1^{\text{floor}} \cdot 30^6. \end{aligned}$$

`matveev_v24x_conditional_summary` is the implication from the still-uninhabited `matveev_inequality_real_target`. `bugeaud_v24x_final_link` restates the conditional LLL shape. This is the honest final of

v24.x, not v25: `baker_bound_gap3` stays a def Prop until Mathlib has Matveev 2000 Thm 1.4. The closing software record is [1].

18. WHAT REMAINS

15/16 of $B \leq 2M$ is eliminated by mod-16 residues. The two displayed named rows [196, 1500003] are eliminated that way (`full_honest`). All 352 named $B \equiv 14$ rows are eliminated by fourth-power non-residues (`beal_4_13_13_gap3_B_le_2M_eliminated_B14_full`). Those 86 that survive mod 53 still have an Euler a_{53} in $\{-10, -2, 1, 6, 14\}$ that misses 0, 12 under the Beal equation (`beal_frey_a53_miss_86`); that pack is not Ribet. `conductor_86` and `level_lowering_86` stay Props. `B14_honest` stays the uninhabited pack $\forall$. `Classical.choice` is not BCDT. The old `beal_4_13_13_gap3_B_le_2M_eliminated` stays `Classical.em`. Density from the named-row count to ∞ , full modular contradiction, and Beal $\forall$ still open. $B > 2M$ is the Baker/Bugeaud Prop `baker_bound_gap3`, not a Zsigmondy theorem. Track beyond 2M Step61+ continues the same `smallZsigPrimes` pattern and does not close the Baker bound.

19. WORKING v33 PAPER - BOUNDED QUOTIENT-REES GLUING (INCORPORATED FROM REPLIT STANDALONE)

The original working source was `papers/main.tex`. This section retains its full body, not a truncation. The expanded manuscript is copied to `papers/main.tex` and `paper/v33-standalone/main.tex`, with their PDFs. These working copies are not the frozen deposit. It is a bounded description of the quotient-Rees three-open gluing and the actual blow-up's generic-fibre comparison. It is not a proof of the general Beal conjecture, a global identification of the actual special-fibre pullback with the quotient Rees Proj, or a degree-one / nonvanishing certificate for $\overline{2t}$.

The archived manuscript's deposit-planning language is historical. The current repository audit records a published v33 GitHub release and version DOI 10.5281/zenodo.23054568; see `Beal/AUDIT_v33_HONESTY.md`, lines 79–88 and 120–124. This revision mints no DOI. The current cited chart version remains v37-true-chart, [2]; the old deposit language does not replace that citation. The level-26 foundations work mirror is not the v33 release source. See `Beal/AUDIT_v33_HONESTY.md`, `Beal/ZENODO_DEPOSIT_v33.md`, and `Beal/RELEASE_NOTES_v33.md`. This incorporation preserves the `beal-v38 EQUIV:3 chain ddfb2642 → e466e5a → 792b3f8`. The freeze tag is `beal-v38-freeze-Y-axis-glue`,

with object 437b4c85 at c48b1bd3. The v37 citation remains 10.5281/zenodo.23120540.

19.1. Abstract of the v33 manuscript. The following abstract is retained from the archived working manuscript. Its release and web-hook statements describe that manuscript’s historical planning stage, not the current deposit status recorded above.

This working v33 manuscript records a three-open `OpenCover` of the *quotient* Rees scheme $\text{Proj}(\text{ReesMod2})$ by $D_+(\overline{2t})$, $D_+(\overline{Xt})$, and $D_+(\overline{Yt})$, with affine restriction squares for every ordered pair over abstract product-open pullbacks. `CompatChart2t` (191 lines) identifies the principal divided-chart kernel (F) and proves `chart_eq_quotient_2t`; `CompatChartXt` and `CompatChartYt` (105 lines each) prove $J_X = (G_X : X^\infty)$ and $J_Y = (G_Y : Y^\infty)$ and their quotient-induced chart identifications. Six directed polynomial-to-abstract restrictions satisfy $\text{res}_{\text{poly}} = \text{res}_{\text{actual}} \circ \psi_\bullet$. `SpecialFibreGluing` (167 lines) checks six composed `Spec.map` compatibilities and the abstract triple cocycle, giving

$$\text{Proj}(\text{ReesMod2}) \cong \text{Glue}(S_{2t}/(2), S_{Xt}/(2), S_{Yt}/(2)).$$

Product overlaps are abstract pullbacks, not polynomial presentations. For $y^2 + xy = x^3 + 8$, the reduced $2t$ chart is $V^2 + UV = 0$; the point $U = V = 0$ lies only in $D_+(\overline{2t})$ (an illustration, not a separately named Lean theorem). `TateEvenBranchGenericFibre` (59 lines) packages the *actual* blow-up comparison $\text{Bl}_I[1/2] \cong \text{Spec}(R_Z[1/2])$ with $I[1/2] = \mathbb{T}$; the existing 2,249-line chart file is untouched. The seven targeted `lake env lean` direct checks previously passed with printed axioms only `propext`, `Classical.choice`, and `Quot.sound`; the scoped `sorry` scan and `git diff --check` are clean. A standalone direct check of the large special-fibre file timed out without a diagnostic and a full `lake build BealGeneral` remains unverified. We do *not* claim a global isomorphism from the actual special-fibre pullback to the quotient Rees `Proj`, or a degree-one quotient formula or nonvanishing theorem for $\overline{2t}$. This is not a general Beal proof. The intended release and Zenodo source is `beal-conjecture`, not its level-26 work mirror; its active GitHub-side Zenodo `release` webhook does not yet establish delivery, a deposit, or a DOI. A DOI may be minted only after an authorized GitHub Release v33 and subsequent Zenodo verification.

19.2. Scope and charts. Write $R_Z = \mathbb{Z}_2[X, Y]/(f)$ for the translated even-branch surface under the stated nodal and coefficient-divisibility hypotheses, $I = (2, X, Y)$, and $\text{ReesMod2} = \text{Rees}(I)/(2)$, where (2) is generated by the degree-zero base scalar, *not* by $2t$. The integral divided chart has $f(2U, 2V) = 4F$; its principal relation is (F) . The

Xt and Yt charts use saturated graph ideals $(G_X : X^\infty)$ and $(G_Y : Y^\infty)$ rather than the unsaturated graph equations. Quotient-induced chart maps connect these integral calculations to the three reduced chart spectra. These assertions concern the charts under their hypotheses, not an unconditional formula for every Weierstrass model.

19.3. The checked gluing boundary. The three reduced generator opens cover $\text{Proj}(\text{ReesMod2})$. Ordered affine restriction squares live over product opens; the six distinct directed pairs obey the polynomial-to-abstract restriction identities. The polynomial spectra are literal cover objects, but their *overlaps* are abstract product pullbacks. Six Spec.map squares and a triple cocycle yield the global isomorphism displayed in the abstract. No polynomial ring presentation of a product overlap is supplied or inferred.

The illustrative $y^2 + xy = x^3 + 8$ specialization has $f(2U, 2V) = 4(V^2 + UV - 2U^3 - 2)$, so its reduced divided-chart equation is $V^2 + UV = 0$. The point $(U, V) = (0, 0)$ gives direction $[2t : Xt : Yt] = [1 : 0 : 0]$ and belongs to the $2t$ open alone; this is an example, not a global classification theorem.

19.4. Generic fibre and unresolved special-fibre comparison. After inverting 2, the centre I becomes the unit ideal. The packaged generic-fibre isomorphism concerns the *actual* blow-up and the localized translated surface over $D(2)$; it does not identify an integral chart with its generic fibre. In contrast, the checked special-fibre gluing concerns the *quotient Rees Proj*. Chartwise base-change comparisons do not yet give an isomorphism of that Proj with the actual scheme-theoretic special-fibre pullback. Nor is $(\text{ReesMod2})_1 \cong (I/2I)t$ or $\overline{2t} \neq 0$ certified here. These missing steps are not used as premises of the quotient- Proj gluing.

19.5. Verification and deposit route. The seven direct Lean checks and their axiom reports are recorded in `Beal/RELEASE_NOTES_v33.md` and `Beal/AUDIT_v33_HONESTY.md`. They are not a verified full-target build: the standalone large special-fibre source check timed out, although the passing modules imported its compiled declarations. The existing `TateEvenBranch.lean` (2,249 lines) passed its separate direct check and was not refactored.

`beal-conjecture` is the whole-project main audit and intended GitHub-release source. The level-26 scaffold's Lean evidence is cited through the vendored `Level26/BealLevel26Foundations/lean/` tree at the immutable `beal-conjecture` baseline, not through an external code pointer. Keep that scaffold separate from an inhabited global

Beal result. The archived manuscript said, “Neither a tag nor a release nor a v33 DOI exists yet.” That sentence is historical: the current audit records the v33 release and DOI. This revision performs no release, deposit, DOI mint, or mirror push.

19.6. PDF printing. Build from `paper/` with `latexmk -pdf mcom-draft.tex` or `tectonic mcom-draft.tex`. The generated file is `paper/mcom-draft.pdf`. The single source directory is `paper/`.

20. IN-DEPTH STUDY OF THE THREE PROPS: PROBLEM AND 2-YEAR RESOLUTION PLAN

This is a research programme, not new proved mathematics or a \$1M Beal Prize claim. The submission concerns a bounded result. A two-year horizon is a planning interval, not a guarantee of resolution.

20.1. Three gap propositions are not the three research axioms. The study areas are `baker_bound_gap3`, `conductor_86`, and `level_lowering_86`. The uninhabited `B14_honest` pack is related. A proposition definition asserts nothing; an axiom supplies an assumption to dependent proofs. The three named research axioms are different:

- (1) `darmon_merel_4413_axiom`, in `BealTrueV25.lean`, assumes the named coprime $(4, 4, 13)$ no-solution proposition. Its source name is not a verification of a bibliographic attribution or a formal proof of that statement.
- (2) `frey_modular_13`, in the vendored `Modularity/FreyModularity_13.lean`, assumes modularity of a displayed coefficient-record type. The predicate also has a displayed constructor: proving that model-level predicate is not proving BCDT for an elliptic curve over $\mathbb{Q}$.
- (3) `ribet_level_lowering_26`, in the vendored `Modularity/RibetLevelLowering_26.lean`, assumes

$$\text{frey_conductor_26} = 26 \implies (\text{ExistsNoncuspidal_26} \implies \text{False}).$$

The conductor is a displayed label and the conclusion uses the displayed point type. Neither a Selmer certificate nor this assumption is a formal Ribet theorem about a residual representation.

Replace mathematical gaps, not merely the spelling axiom.

20.2. Prop 1: the exact Baker-bound interface. The vendored `BealGap3BakerUpperBound.lean` defines `baker_bound_gap3` using a witness $B_0 = 10^6$ and

$$\forall B > B_0, \quad \neg \exists A \in \mathbb{N}, \quad A^4 + B^4 = (B + 3)^{13}.$$

The older beyond-2M range is not this declaration’s cutoff. The equation is $(4, 4, 13)$, not $A^4 + B^{13} = C^{13}$. `hBaker` and `fullB0Search` remain separate inputs. Neither the 62500-value cover through 10^6 nor `Step61+smallZsigPrimes` witnesses supply an above-cutoff bound.

20.3. Analytic obligations and proposed milestones. A Matveev-based argument needs an exact theorem statement, nonzero logarithmic form, algebraicity and positivity hypotheses, degree and logarithmic-height bounds, explicit constants, and an effective reduction at the intended cutoff. A schematic exponential lower bound alone does not force $B \leq 10^6$ or $B \leq 2M$: the heights and coefficient bounds may themselves depend on B . The existing conditional logarithmic calculations must be connected to those hypotheses, with all inequalities in the required direction.

- Months 0–6: audit the pinned height and logarithm APIs; choose a precise published Matveev formulation and a compatible height normalization. Establish the needed elementary height identities.
- Months 6–12: attempt the missing lower-bound formalization, recording any theorem imported as an explicit assumption rather than announcing a proved Matveev theorem.
- Months 12–18: derive and verify the application-specific constants and reduction. The acceptance test is an actual proof of the exact Baker proposition, with no new research axiom.
- Months 18–24: consolidate reusable results and, where suitable, prepare an upstream Mathlib contribution. Upstream acceptance is not a logical prerequisite if the proved result remains vendored.

The v24.x closing software record [1] remains conditional. A version label cannot substitute for the missing theorem.

20.4. Prop 2: conductor study, not a conductor label. In `BealFrey B14.lean`, `conductor_86` is an alias of `FreyModularity.level_lowering_to_26`; it is not a definition of an actual conductor function or the assertion $N(E) = 26$. The source explicitly disclaims the old stronger $N \mid 2^5 \cdot 3 \cdot 13$ claim. The example $63982 = 2 \cdot 31991$ prevents replacing the recorded radical by 3. The packed bound $2^5 \text{ rad} \cdot 13$ is not yet a comparison with a genuine global conductor.

The required work starts with the actual Beal Frey cubic $y^2 = x(x - A^4)(x + B^4)$, nonsingularity, an integral Weierstrass model, and justified changes to minimal models. Local analysis must cover every relevant prime, including bad characteristic and possible wild contributions. One may not simply enumerate 86 values of B : the intended curves also depend on A .

Months 0–3 target model and valuation interfaces; months 3–6 target verified local case analyses; months 6–12 target the comparison with a defined global conductor and any admissible level-lowering level. The outcome may require a corrected, separately named proposition rather than a proof of an unsuitable historical placeholder. The finite `J0_26_Decom.lean` records do not prove a Jacobian decomposition.

20.5. Prop 3: level lowering and the honest finite pack. `level_lowering_86` is another alias of the displayed level-lowering proposition. A genuine replacement requires a modular elliptic curve, its residual representation, irreducibility, local hypotheses, an actual level-lowering theorem, and transport to the relevant newform coefficients. An integer trace miss is not a proof of those inputs.

The all-cell Euler table includes 0 at the singular cell. Under the equation, the nonsingular values $\{-10, -2, 1, 6, 14\}$ give the stated miss of 0, 12 modulo 13. The displayed-cubic traces $-2, 24$ and the locked level-26 traces must not be interchanged without a model and prime-specific comparison. The two cubics remain separate.

The 352 named rows are already covered by the independent $266 + 77 + 9$ residue argument. This does not inhabit the differently typed `B14_honest` modular pack. Rewriting an enumeration cannot create missing modularity or make a false interface true. `Classical.choice` is not BCDT, and `Classical.em` proves a disjunction, not its desired branch.

Window	Workstream	Evidence required for acceptance
0–2 months	Finite interface audit	Exact populations and kernel-checkable residue proofs; no transfer to the modular pack by renaming.
2–12 months	Residual representations	Actual representation and local hypotheses; explicit missing modularity and level-lowering inputs.
12–24 months	Newform comparison	Proved level and coefficient transport, followed by a genuine contradiction under the stated hypotheses.

20.6. Two-year review gate and limits. The analytic, conductor, and representation workstreams may proceed in parallel, but their mathematical dependencies determine what can be combined. At each six-month review, record proved declarations, open hypotheses, corrected interfaces, and unsuccessful approaches. Do not replace an unresolved assumption with another opaque assumption.

At month 24, a merge is justified only for the proofs actually completed. If a required bridge remains open, retain it in the statement. There is no promise that all three propositions will be resolved. Even successful completion would concern the precisely stated gap-three programme, not all exponent triples of Beal’s conjecture. Density beyond named populations, the general modular inference, and unrestricted Beal remain open. No prize claim is made.

21. FORMALIZATION, SCAFFOLDING AND OVERCOMING MATHLIB GAPS

21.1. Repository-only provenance and truthful interfaces. Project-owned Lean evidence is confined to this repository, including the vendored Level26 package. Standard Lean and Mathlib dependencies are acknowledged; unrelated research and private certificates are not proof premises. The immutable baseline 02728795 is a Git commit, not a DOI. This revision creates no release or deposit.

Every claimed result needs four separate records: its exact Lean type, an inhabitant of that type, the inhabitant's axiom dependencies, and its interpretation as the intended mathematics. A finite record may certify coefficients or a residue table, but not automatically a modular curve, Jacobian, L-function, or residual representation. An interface with a proof-valued field is useful only once that field has been constructed without the disputed assumption.

21.2. Existing chart mathematics is not merely finite data. The even-branch work includes genuine algebraic and scheme-theoretic constructions; it must not be described wholesale as finite records. The divided chart, saturated graph ideals, chart restrictions, and quotient-Rees cover are distinct objects. For $I = (2, X, Y)$, the reduction $\text{Rees}(I)/(2)$ is by the degree-zero base scalar, not by $2t$. The v33 result concerns the quotient-Rees three-open gluing and the actual blow-up's generic-fibre comparison.

The product-open overlaps are abstract pullbacks, not explicit polynomial overlap presentations. Neither a global identification of the actual special-fibre pullback with the quotient Rees Proj nor degree-one/nonvanishing of $\overline{2t}$ is asserted. The large special-fibre direct-check file is untouched. There is no general Beal proof in this geometric packaging.

21.3. From scaffolding to mathematical proof. Begin with a specification and a precisely typed gap, not a placeholder disguised as a theorem. Prove elementary components, transport them through explicit comparison maps, and only then construct the proof-valued field of the downstream interface. Retain the proposition definition and add its proof; a proposition does not cease to be a proposition when proved. If a model predicate must be replaced, document the change and prove any required equivalence rather than silently strengthening the meaning of an old name.

21.4. Matveev example: the bridge to be constructed. On a positive solution of the actual equation, put $C = B + 3$. The logarithmic form compatible with the existing calculations is

$$\Lambda = 13 \log C - 4 \log A = \log \left(1 + \frac{B^4}{A^4} \right).$$

The positivity hypotheses justify the divisions and logarithms. The elementary upper estimate is

$$0 < \Lambda \leq \frac{B^4}{A^4}.$$

This does not provide a lower bound. A Matveev application must establish a theorem-specific lower estimate for the same nonzero form, with degree, heights, logarithm choices, and coefficient parameters proved rather than supplied as certificate labels. Different formulations, including multiplicative forms, require their own transport lemmas.

Suggested new work belongs beside the existing `Beal/MathlibMissing/` infrastructure or in a clearly named analytic package. A height module and a Matveev module are proposed deliverables, not files implemented by this manuscript. No fixed line count is a measure of completeness.

- (1) Define the exact height normalization and domain; prove the algebraic-number and elementary height properties needed for the intended application.
- (2) Formalize the precise lower-bound theorem, or expose its missing proof as a hypothesis. Check every constant and nonvanishing condition.
- (3) Compare the upper and lower estimates. Prove any further effective reduction needed to reach $B_0 = 10^6$; do not assume that the displayed inequalities alone yield that cutoff.
- (4) Supply the exact finite-search proof as a separate argument. Only a proved beyond-cutoff bound together with an exhaustive matching finite cover can close the unrestricted gap-three statement under its other hypotheses.

The named 352-row cover through 2M and the 62500-value cover through 10^6 are different populations. The input `BealFinalData.fullB0Search` and the premise `hBaker` remain explicit. The general common-prime theorem is itself an input to the final forward wrapper; projecting that field does not construct the theorem. Likewise, replacing the named Darmon–Merel axiom by a displayed data constructor would not prove its number-theoretic conclusion.

21.5. Tate, Ribet, and finite checking. A proposed Tate module must connect discriminant and c_4, c_6 valuations, integral changes of variables, minimality, reduction type, and local conductor exponents. The existing local and chart lemmas should be reused at their precise hypotheses. A computed valuation alone does not determine reduction type or a global conductor, especially at 2.

A proposed level-lowering module must refer to actual elliptic curves and Galois representations. It must justify the lowered level and compare the coefficients of the correct newforms at good primes. The computational labels `26a1`, `26b1`, their finite prefixes, and the certificate `ribet_secured_by_certs` do not construct those objects or prove Ribet’s theorem. Removing the displayed `frey_modular_13` assumption by using its model predicate’s constructor is not a proof of modularity over $\mathbb{Q}$.

For finite statements, prefer kernel-checkable `decide` and arithmetic proofs when feasible. A use of `native_decide` must be audited for its trusted computational dependencies; it is not automatically compatible with a classical-trio-only axiom policy. Eliminating choice in a finite witness is optional proof engineering, not the missing BCDT argument.

21.6. Verification and publication gates.

- (1) Build the exact active Lean roots. A lexical `sorry/admit` scan is separate from a kernel check; neither a cache download nor unrelated green CI proves a target.
- (2) Print axioms for each submitted declaration. The three research axioms are assessed per theorem, not attributed to every chart result by association.
- (3) Keep the recorded seven-declaration v38 check distinct from fresh full-build status. Historical reports and timeouts remain labelled as such in the source ledger.
- (4) Compile all three manuscript paths, check 26 searchable pages, undefined references, overfull boxes, physical-page overflow, and zero-width hyperlink borders. A PDF check says nothing about completing a missing Lean theorem.

The chain `ddfb2642` $\rightarrow$ `e466e5a` $\rightarrow$ `792b3f8` and the frozen tag objects are unchanged. The v37 citation remains `10.5281/zenodo.23120540`; the published v33 version is `10.5281/zenodo.23054568`. No DOI is minted. This is a bounded manuscript with a conditional research programme, not a completed general proof or a \$1M claim.

22. GENERAL BEAL STATEMENT AND THE v38 TRUE-CHART CERTIFICATE

In this section Y, S denote the cusp-chart variables, independently of the coefficient-ledger notation in section 1. Write $\overline{N} = \text{closedYtToCusp } N$ for the cusp class.

Theorem 2 (True-chart presentation and E_i certificate (v38)). *Let*

$$R = \mathbb{F}_2[a, b][X, Y, U, V],$$

$$I_{\text{true}} = (XU, Y - XV, Y^2 - X^3, U + X^2 + XV^2).$$

The true model R/I_{true} is ring-isomorphic to the fixed chart `chart_Dplus_Xt_ring valuationOneCurve 0 0` via `chart_Dplus_Xt_true_presentation`. Its map $\iota = \text{chartOfModelTrue}$ is injective. For $N : \text{modelYtChart_fixed}$ with `chartOfModelTrueY\_fixed` $N = 0$, the certificate supplies a cusp representative

$$\overline{N} = A + YB + Y^2C + Y^3(E_0 + SE_1 + S^2E_2).$$

Here A, B, C have Y -degree zero, and E_0, E_1, E_2 have S -degree zero. The high part $Y^3(E_0 + SE_1 + S^2E_2)$ has S -degree at most two. Its chart image equals that of the low part $A + YB + Y^2C$; writing the high-part image as H_N , it satisfies $(1 + YS^3)H_N = 0$. The recorded nonvanishing and annihilator conjuncts are retained: no vanishing of B or E_i is inferred.

Proof. The ring presentation is `RingEquiv.ofBijective` applied to the existing injectivity and surjectivity proofs in `Beal/MathlibMissing/ChartInjective.lean`, lines 3162–3176. The v38 conjunction in `Beal/MathlibMissing/ChartTrueEquiv.lean`, lines 261–338, assembles `chartOfModelTrue_injective`, `chartTrueEquiv_inj_from_Ei_constraint N hN`, and `chartTrueEquiv_annihilator_presentations`. The statement above displays their chart-presentation and normal-form components, not a new arithmetic implication. $\square$

Nonvanishing and overlap invariants. The $B = 1$ example remains nonzero outside the cusp; Y^3 is nonzero and outside the cusp ideal. In the centre-power assertion, I is the centre ideal of the fixed 2-adic model, not the characteristic-two ideal I_{true} . In particular,

$$\text{centreNormalPoly}(X^3 - 1)0 \notin I^2, \quad \text{centreAlphaBound } 20 = 1.$$

The constant coefficient of $X^3 - 1$ is a unit in the 2-adic coefficient ring, incompatible with that required centre-power bound. The relation $X^2(X + V^2)z = 0$ is distinct from $B = 0$; $X + V^2$ stays outside the cusp ideal with nonzero image. On the overlap, $SV = 1$, $Y = XV$, and

Write $\mathcal{M} = \text{modelYtChart_fixed}$, $\mathcal{C} = \text{chartYtCuspRing}$, and $\mathcal{T} = \text{chart_Dplus_Yt_ring valuationOneCurve 0 0}$. The named maps are $c = \text{closedYtToCusp}$, $f = \text{chartOfModelTrueY_fixed}$, and $\varphi = \text{chartOfCuspY} = f \circ \text{cuspYtToClosed}$.

$$\begin{array}{ccc} \mathcal{M} & \xrightarrow{c} & \mathcal{C} \\ f \downarrow & & \downarrow \varphi \\ \mathcal{T} & \xlongequal{\quad} & \mathcal{T} \end{array}$$

The square commutes because $\text{cuspYtToClosed} \circ c = \text{id}_{\mathcal{M}}$. For the kernel hypothesis $f(N) = 0$, write the cusp representative as $\bar{N} = [L] + [Q]$, where

$$\begin{aligned} L &= A + YB + Y^2C, \\ Q &= Y^3(E_0 + SE_1 + S^2E_2), \quad \deg_S Q \leq 2. \end{aligned}$$

Then the certificate gives

$$\varphi([Q]) = \varphi([L]) = H_N, \quad (1 + YS^3)H_N = 0.$$

The image equality uses characteristic two, not coefficient vanishing. The maps and inverse identity are defined in `Beal/MathlibMissing/ChartYt.lean` and `Beal/MathlibMissing/ChartYtPresentation.lean`, lines 47–73 of the latter file.

FIGURE 1. Defined cusp/chart maps and the annihilator presentation. No ring map $D_+(Xt) \rightarrow \text{overlap}$ is asserted; no vanishing of B or E_i is inferred.

$X = YS$. With $\text{ann}(f)$ denoting the annihilator of the principal ideal generated by f , the recorded comparisons are

$$\begin{aligned} \text{ann}(1 + YS^3) &= \text{ann}(X + V^2), \\ \text{ann}(X^2) &= \text{ann}(Y^2), \\ \text{ann}(1 + YS^3) &\neq \text{ann}(X^2). \end{aligned}$$

The distinguishing witness is Y^2 : it annihilates the factor $1 + YS^3$, not the power X^2 . These are annihilator presentations. The source supplies no ring map $D_+(Xt) \rightarrow \text{overlap}$; no such arrow is inferred. The kernel hypothesis on N refers to its defined chart map, not to an invented map into the overlap.

The recorded general statement. This records the general Beal conjecture succinctly: for positive bases and exponents at least three, an equality $x^p + y^q = z^r$ entails a common prime divisor. The formal forward interface has an explicit input $D : \text{BealFinalData}$, including the

field D .bealTheorem.commonPrime:

$$D : \text{BealFinalData}, \quad x, y, z > 0, \quad p, q, r \geq 3, \\ x^p + y^q = z^r \implies \exists \ell \in \mathbb{N}, \quad \ell \text{ prime}, \quad \ell \mid x, \ell \mid y, \ell \mid z.$$

`beal_main_theorem_forward` returns that common-prime input field in `BealFinal/Main.lean`, lines 74–80. Theorem 2 displays chart injectivity and the accompanying E_i and annihilator data; it does not supply the common-prime input. The v33 result in section 19 remains separately bounded.

22.1. Immutable provenance.

Tag or version	Object	Target or citation
<code>beal-v38-freeze-Y-axis-glue</code>	437b4c85	c48b1bd3
<code>beal-v38-equiv-Ei-constraint</code>	c2f530ab	792b3f8
<code>v37-true-chart</code>	10.5281/zenodo.23120540	

The proof record is `ddfb2642` $\rightarrow$ `e466e5a` $\rightarrow$ `792b3f8`: the E_i constraint associated with `025b34c2`, then the nonzero $B = 1$ example, then EQUIV:3. The code baseline is `02728795e15f6c858f0d360d560f80e4efb252a2`; the manuscript revision starts from PR28 at `75b8a3a5`. These commit and tag-object identifiers are not new DOIs.

APPENDIX A. EXACT-TARGET REPRODUCIBILITY AND SOURCE LEDGER

Retain Lean/Mathlib 4.12.0 and the checked-in manifest. Do not run `lake update` for this pinned replay. From the repository root, run each target separately and retain its complete log and actual exit status:

```
lake --version
lake exe cache get
lake build BealMathlibMissing
lake build BealEven
```

After successful target builds, directly check `Beal/MathlibMissing/ChartTrueEquiv.lean`; its seven `#print axioms` commands are at lines 340–346. The repository reports [`propext`, `Classical.choice`, `Quot.sound`] for those declarations. Report any `sorryAx` in the replay. This is scoped, not an axiom certificate for every module. Other modules contain explicit declarations `darmon_merel_4413_axiom`, `frey_modular_13`, and `ribet_level_lowering_26`; their role must be assessed for each cited theorem.

The manuscript review’s independent cache-setup attempts exited 137 before either requested target ran; the environment recorded two OOM kills. A lexical scan found no active `sorry`, `admit`, or `sorryAx` tokens in the resolved local import closures: 57 files for the 18 `BealMa`

`thlibMissing` roots and 39 files for the one `BealEven` root. External dependencies were excluded. That scan is not elaboration or a printed axiom report. The inspected workflow builds other Matveev targets; its status does not certify these two exact targets. The earlier PR28 PDF built to 11 searchable pages; pagination changes in this revision are allowed for readability.

Repository-only Lean evidence. Use immutable paths within `DavidFox998/beal-conjecture`:

- Finite trace data: `Level26/BealLevel26Foundations/lean/BealLevel26Foundations/Beal/FullProof/Level26_Newforms.lean`.
- Named residue cover and conditional bound: `BealFreyMod53Kill.lean`, `BealFreyModQKill.lean`, and `BealGap3BakerUpperBound.lean` in that same `FullProof/` directory.
- Bounded gluing: `Beal/Beal.General/SpecialFibreGluings.lean`.
- True chart and v38 data: `Beal/MathlibMissing/ChartInjective.lean` and `Beal/MathlibMissing/ChartTrueEquiv.lean`.
- Cusp/chart maps in fig. 1: `Beal/MathlibMissing/ChartYt.lean` and `Beal/MathlibMissing/ChartYtPresentation.lean`.
- General common-prime input interface: `BealFinal/Main.lean`.
- Finite coefficient records: `Level26/BealLevel26Foundations/lean/Beal/Foundations/J0_26_Decomposition.lean`.

Standard Lean, Mathlib, and SageMath dependencies remain acknowledged. Repository-only source provenance is not a claim of having no external libraries. A `def Prop`, a folder name, or a certificate label is not by itself an inhabited theorem.

APPENDIX B. VERSION HISTORY: THIS REPOSITORY ONLY

This appendix records versions and their stated scope in `DavidFox998/beal-conjecture`. It introduces no new mathematics, changes no Lean theorem, and mints no tag, release, or DOI. Historical verification reports remain distinct from independent replay.

`v24.4.0`: historical, superseded. The record at `db7a556`, tagged `v24.4.0-Beal-44-13-Level-26-v24x-Final-Summary`, has version DOI `10.5281/zenodo.22732209` and concept DOI `10.5281/zenodo.22379293`. It links the logarithmic upper bound to a conditional Matveev ratio toward B_0 , not an unconditional Baker bound. Its stated historical scope reports zero `sorry` and `[propext, Classical.choice, Quot.sound]`. That superseded report is not a fresh replay here; the keyed bibliography record is [1].

v33: bounded gluing and actual generic fibre. `Beal/Beal.General/SpecialFibreGluing.lean` records quotient-Rees three-open gluing, alongside the actual blow-up’s generic-fibre comparison. The scope is bounded: not general Beal, not a global special-fibre identification, and not a degree-one or nonvanishing certificate for $\overline{2t}$. `Beal/RELEASE_NOTES_v33.md` and `Beal/AUDIT_v33_HONESTY.md` supply the release and audit record. The standalone check of `TateEvenSpecialFibre.lean` timed out; the passing v33 modules imported its compiled declarations. The separate `TateEvenBranch.lean` has 2249 physical lines in that audit and a separately reported direct check.

The archived manuscript formerly at `papers/main.tex` is retained at `paper/v33-standalone/main.tex` and incorporated in Section 19 of PR28. Its sentence “Neither a tag nor a release nor a v33 DOI exists yet” is historical, not present release status. `Beal/AUDIT_v33_HONESTY.md`, lines 79–88 and 120–124, records the v33 release and DOI 10.5281/zenodo.23054568. That historical DOI is not substituted for the current v37 citation.

v37: true-chart presentation. The current chart-version citation is 10.5281/zenodo.23120540, v37-true-chart [2]. The `chart_Dplus_Xt_ring` true-model presentation R/I_{true} is recorded in `Beal/MathlibMissing/ChartInjective.lean`, lines 3162–3176. This is a chart presentation, not a general Beal proof.

v38: EQUIV:3 at 792b3f8. `chartOfModelTrue_injective_from_Ei_constraint` is recorded in `Beal/MathlibMissing/ChartTrueEquiv.lean`, lines 261–338. The chain `ddfb2642` $\rightarrow$ `e466e5a` $\rightarrow$ `792b3f8` records, respectively, the E_i constraint associated with `025b34c2`, the nonzero $B = 1$ example, and the injectivity restatement. The constraint has S -degree at most two and $(1 + YS^3)$ annihilates the high image; $Y^3 \neq 0$ remains outside the cusp ideal. The freeze object is `437b4c85` at `c48b1bd3`; the separate EQUIV object is `c2f530ab` at `792b3f8`. The certificate and its explicit hypotheses remain those of Section 20.

Baseline and manuscript snapshots. The main-source baseline `02728795e15f6c858f0d360d560f80e4efb252a2` contains the PR25 merge `a2a23292`. The pinned tools are Lean 4.12.0 and Mathlib v4.12.0, with the vendored package `Level126/BealLevel126Foundations`. The PR28 manuscript sequence is `c67dc4e` $\rightarrow$ `d637e91` $\rightarrow$ `7531386`. The Phase 4 snapshot at `7531386`, before this appendix, has 17 searchable pages: Tectonic exit zero, no overfull boxes or undefined references, and no detected page-boundary clipping; underfull spacing warnings remain. That page count describes the cited snapshot, not every later build.

Verification boundary. Independent cache setup exited 137 before the two requested targets ran; two OOM kills were recorded. Fresh exact-target builds and printed axiom reports remain pending verification. The local lexical scan covered 57 files from the 18 `BealMathlibMissing` roots and 39 files from the one `BealEven` root, finding no active proof-hole tokens. It excludes external dependencies and is not a kernel check. The scoped repository report for the seven declarations printed in `ChartTrueEquiv.lean` is `[propext, Classical.choice, Quot.sound]`. Other modules declare `darmon_merel_4413_axiom`, `frey_modular_13`, and `ribet_level_lowering_26`; dependencies must be assessed per theorem, not inferred from the chart’s axiom report.

APPENDIX C. VERIFICATION STATUS FOR EDITING: FINAL CHECKLIST

- The chart axiom footprint `[propext, Classical.choice, Quot.sound]` is a scoped repository report. Independent replay remains unverified.
- The lexical scan found no active proof-hole tokens in 57 local `BealMathlibMissing` files and 39 local `BealEven` files. External dependency sources were excluded; this is not a substitute for a kernel build.
- Historical exit-zero reports for those two targets were not reproduced here. Independent cache setup exited 137, with two OOM kills before either requested target ran.
- PR28 is a manuscript branch, not audited main. The 17-page build at 7531386 and 18-page build at 621d106 are distinct snapshots; neither page count is silently assigned to the other commit or to a later revision. PDF layout checks do not certify Lean proofs.
- `conductor_86` and `level_lowering_86` remain recorded as uninhabited Props; `B14_honest` is the uninhabited universal pack, not a BCDT theorem. `Classical.choice` supplies neither Wiles/BCDT nor Ribet. `beal_4_13_13_gap3_Ble_2M_eliminated` proves excluded middle for a witness predicate using `Classical.em`; it does not prove Beal elimination. Infinite density, full modular contradiction, and unrestricted Beal remain open. Beyond two million, the `baker_bound_gap3` premise is not supplied by Zsigmondy or the Step61+ `smallZsigPrimes` continuation.

- The complete archived `papers/main.tex` body remains incorporated in Section 19 with its bounded v33 disclaimers, separate from the v38 certificate and general common-prime input.
- v33 glues the quotient-Rees `Proj(ReesMod2)` using abstract product pullbacks, not polynomial product-overlap presentations. It gives neither a global actual special-fibre identification nor a degree-one/nonvanishing certificate for $\overline{2t}$. The v38 true-chart/ E_i certificate is not an unrestricted Beal proof.
- The current chart citation remains `10.5281/zenodo.23120540.02728795` is a commit, not a DOI. This editing checklist performs no submission, merge, release, or DOI mint.
- No Lean sources, dependency pins, archived `paper/v33-standalone/` files, tags, releases, or DOIs are changed by this manuscript follow-up.

REFERENCES

- [1] DAVID FOX, *Beal Level 26 Foundations v24.4.0: v24.x Final Summary Linking Log Upper Bound to Conditional Matveev Ratio Toward B_0* . On gap-3 $A^4 + B^4 = (B + 3)^{13}$, $|\Lambda| = \log(1 + B^4/A^4) \leq B^4/A^4 = B^4/((B + 3)^{13} - B^4)$ with $0 < \exp(C_{\text{exp_bound}}) < 10^{-12} < 1$, $C_{\text{exp_bound}} = -\text{height}_{B_0} \log \text{height}_{B_0}$, $\text{height}_{B_0} = 104382751019310000000 = C_1^{\text{floor}} \cdot 30^6$. IF Matveev (2000) Thm 1.4 target $|\Lambda| > \exp(C)$ held, THEN $0 < \exp(C) < B^4/A^4$ and $B^4(1 + \exp(C)) > \exp(C)(B + 3)^{13}$ toward Baker $B_0 = 10^6$. No lower bound claimed. The historical release record reports Lean verification, 0 sorry, and `[propxt, Classical.choice, Quot.sound]` for its stated scope; this is not a fresh verification result here. DOI <https://doi.org/10.5281/zenodo.22732209>, concept <https://doi.org/10.5281/zenodo.22379293>, tag `v24.4.0-Beal-44-13-Level-26-v24x-Final-Summary` on `db7a556`.
- [2] DAVID FOX, *Beal conjecture repository: v37-true-chart*. Version DOI `10.5281/zenodo.23120540`. The true-chart presentation is cited through this chart-version record and the immutable repository paths above, not as a general Beal proof certificate.